



Khawaja Yunus Ali University

Assignment

Name of the Department: Computer Science and Engineering

Course Code: CSE 0713-1103

Course Title: Electrical Circuit

Assignment No.: 01

Name of the Assignment : Calculation of Electricity bill of household .

Date of Submission : 05-02-2025

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Instructor Signature & Date

Submitted by –

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Batch No: 18th

Semester: 1st Year 2nd Semester

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Topic: Calculation of Electricity bill of household.

Step 1: The number of various equipment's available in my household.

- i) Light: 3
- ii) Refrigerator: 1
- iii) computer : 1
- iv) Water Pump : 1

Step 2: The power ratings of every equipment in my household.

- i) Light: 10w (each)
- ii) Refrigerator : 120 W
- iii) computer: 200W
- iv) Water Pump : 400W

Step 3: For each equipment, estimate the amount of time used.

- 1. Light: 7 hour per day.
- 2. Water Pump : 40 minute par day. = 0.667 hours
- 3. Refrigerator: : Chargers for 4 hours daily.
- 4. computer: 2 hour per day.

Step 4: calculate total energy consumed (in unit)

$$\text{Energy (kwh)} = \frac{\text{Power (W)} \times \text{Time (h)} \times \text{number}}{1000}$$

For, 3 light

$$\begin{aligned} &= \frac{3 \times 7 \times 10}{1000} \times 30 \\ &= 7.2 \text{ unit} \end{aligned}$$

For, 1 refrigerator

$$\begin{aligned} &= \frac{1 \times 120 \times 6}{1000} \times 30 \\ &= 14.4 \text{ unit} \end{aligned}$$

For, 1 computer

$$\begin{aligned} &= \frac{1 \times 2 \times 200}{1000} \times 30 \\ &= 12 \text{ unit} \end{aligned}$$

For ,1 water pump

$$= \frac{1 \times 0.667 \times 400}{1000} \times 30$$
$$= 8 \text{ unit}$$

Total energy consumed = (7.2+14.4+12+8) = 41.6 unit

Step 5:

0-75 unit = 4.63 taka

75-200 unit = 6.85 taka

Total cost of unit and other costs- Total cost of unit=(41.6× 4.63) =
192.60 taka

Meter rent = 10 taka

Demand charge = 40 taka

Vat(5%)= (192.60 × 5%) = 9.63 taka

Step 6:

Total electricity bill-

Total electricity bill of august = (192.61+10+40+9.63)=252.23 taka

Step 7:

Compare this bill with the bill of paper- Electricity bill of January on paper of bill = 249 taka

Compare = $(252.23 - 249) = 3.23$ taka

Step 8:

The reason for variation-

1. Estimation Errors: The estimated usage times or power ratings might not match the actual usage.
2. Standby Power Consumption: Many devices consume power even when turned off, which might not have been included.
3. Tariff Variations: The actual billing might include additional charges or different rates that weren't considered in the calculation.
4. Meter Accuracy: Differences between the household's meter readings and the utility company's recorded consumption could cause discrepancies